

Practice Quiz: Agents — The Pattern-Reader's Mind

Part A: Multiple Choice

1. In Active Inference, a pattern-reading crocheter is best described as: A) A passive receiver of instructions B) An inference agent who generates predictions and updates them based on observations C) A machine that executes commands without interpretation D) A random decision-maker
2. When an experienced crocheter reads "dc in each st across" and instantly visualizes a row of tall, open stitches, this is an example of: A) Memorization without understanding B) The generative model producing a prediction before the action occurs C) Following instructions mechanically D) Guessing randomly about the outcome
3. The instruction "work in pattern as established" requires the agent to: A) Look up the pattern in a reference book B) Have extracted and internalized the pattern logic from previous rows C) Start over from the beginning D) Skip that row entirely
4. A beginner experiences more prediction errors than an expert because: A) Beginners are less careful B) The beginner's generative model is sparser and less accurate C) Patterns are written only for experts D) Beginners use cheaper yarn
5. Which of the following is an **active state** of the pattern-reading agent? A) The written text of the pattern B) The photo of the finished project C) Deciding to frog and redo a section that does not match expectations D) The gauge listed in the pattern header
6. When you form a mental image of the finished piece before starting, you are: A) Wasting time you could spend stitching B) Running a simulation with your generative model to predict the posterior C) Ignoring the pattern's instructions D) Copying someone else's work
7. Two experienced crocheters may interpret the same ambiguous pattern instruction differently because: A) One of them is wrong B) Their generative models contain

different priors from different experiences C) Only one of them can read D) Patterns have only one possible interpretation

Part B: Short Answer

1. Describe the prediction-observation-update loop as it plays out during a single row of crochet. What prediction does the agent make, what observation follows, and how does the agent respond to a mismatch?
2. Explain why a pattern with no finished-object photo is harder for a beginning agent to work from than one with detailed photos. Use the concept of generative model richness.
3. You are an experienced crocheter who has just picked up a complex lace pattern in a style you have never tried before. In what ways are you simultaneously an expert agent and a novice agent? How does your model handle this mix?